

Tensile and bonding behaviours of hybridized BFRP–steel bars as concrete reinforcement

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HIGHLIGHTS

- A hybridized basalt fibre–reinforced polymer and steel (BFRP–steel) rebar is proposed in this study.
- The proposed method aims at providing a possible anticorrosive reinforcement for concrete components.
- The hybridized BFRP–steel bar enabled a more balanced tensile behaviour between steel and FRP reinforcements.
- The bond strength of the proposed hybridized bar was investigated based on four types of surface treatments.

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ABSTRACT

Corrosion of steel increases the cost of maintenance and reduces the long-term performance of steel-reinforced concrete structures. Fibre reinforced polymer (FRP) bars have been developed as promising alternatives. However, the application of FRP bars is limited by high initial cost, low elastic modulus, and brittleness. Therefore, a hybridized basalt FRP and steel (BFRP–steel) rebar is proposed to provide another possible anticorrosive reinforcement for concrete. The proposed rebar protects the steel core with BFRP wrap, and the low cost of steel improves the feasibility of using fibre reinforcement by significantly reducing the total material cost. Four BFRP–steel and seven glass FRP bars were experimentally tested to investigate their tensile strength. Results showed that the hybridized bars allowed a more balanced tensile behaviour, which not only improved the ultimate tensile strength by 47% compared with steel reinforcement, but also overcame the shortage of FRP bar's low elastic modulus by introducing a 169% improvement. Meanwhile, the bond strengths of the FRP bars was tested by pullout tests of four groups of surface treatments. Results indicated that the sand-coating treatment had the highest level of bonding capacity but had a brittle failure mode. A more ductile but less strong bonding behaviour could be achieved by combining the surface treatment of the sand-coating with spiral-wounding.

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1. Introduction

Steel-reinforced concrete is one of the most widely used construction materials around the world based on its high tensile and compressive strength provided by the combination of concrete and steel rebar. However, the long-term performance of steel-reinforced concrete is reduced because of the corrosion of steel; and high costs of inspection, maintenance, and restoration of corroded steel reinforcements are required. Therefore, fibre-

reinforced polymer (FRP) reinforcements are extensively studied to seek the probability and feasibility of replacing steel rebar, which includes research on mechanical properties of concrete structures reinforced by glass FRP (GFRP) [3,2,26,23], carbon FRP (CFRP) [22,29,20,19], and basalt FRP (BFRP) [9,12,13,5,11,16].

Recently, the BFRP bar has been developed as a promising alternative to traditional FRP bars for reinforcing concrete structures because it behaves more balanced between mechanical properties and costs than GFRP and CFRP bars. Lopresto et al. [21] found that the BFRP has higher compressive strength, Young's modulus, and flexural strength than the E-glass FRP. When compared with S-glass and carbon fibre, the basalt fibre shows lower mechanical properties. However, S-glass and carbon fibre are much more

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expensive than basalt fibre, which makes basalt fibre more applicable.

Therefore, more and more studies have been conducted on BFRP reinforcement because of its exceptional properties and relatively low costs. Gopinath et al. [18] stated that the load-carrying capacity could be doubled without losing ductility when BFRP bars were applied for the near-surface mounted strengthening of RC beams. Zhang et al. [34] experimentally and numerically studied the flexural deflections of concrete beams reinforced with BFRP bars and proposed a new equation to calculate the deflection of the BFRP-reinforced RC beams. Tomlinson and Fam [32] evaluated the flexural and shear performances of concrete beams reinforced with BFRP bars and stirrups, and a higher flexural capacity at rupture was observed when stirrups were applied. Cai et al. [6] executed numerical analyses of BFRP-reinforced engineered cementitious composites (ECC) and concluded that the application of BFRP bars in the ECC matrix is much more efficient than those in conventional concretes. Chen et al. [8] proposed a U-shape BFRP sheet as RC beam reinforcements and found a significant increase of load carrying capacity.

However, compared with steel, the application of FRP reinforcement is limited by its high initial cost, low elastic modulus, and brittleness. Thus, various types of hybrid reinforcements of FRP and steel have been developed by researchers. Qu et al. [25] experimentally and theoretically investigated the load-deflection behaviour of concrete beams reinforced with hybrid GFRP and steel bars. Seo et al. [27] proposed a GFRP steel hybridized rebar and found an approximately 250% improvement of the elastic modulus. Cao et al. [7] developed a hybrid CFRP enclosure and steel rebar reinforcement for an expansive concrete, and superior performance in both mechanical properties and ductility was obtained. Qin et al. [24] numerically studied hybrid FRP-reinforced concrete beams and found that a hybrid reinforcement ratio of FRP and steel should be designed between 1 and 2.5. Ge et al. [17] tested the flexural behaviour of hybrid concrete beams reinforced with both BFRP and steel bars, and results showed that sufficient ductility could be achieved.

Meanwhile, bond strength is another fundamental aspect of BFRP behaviour as the bond property affects the serviceability, ductility and capacity of concrete structures. Studies have been conducted to investigate the durability and bonding strength of using FRP bars as reinforcements for concrete structures. Yan et al. [33] reviewed 682 pullout tests of GFRP bars and determined critical factors that affect bond behaviours, including concrete compressive strength, concrete cover, bar size, embedment length, etc. El Refai et al. [15] experimentally tested the bond behaviour of BFRP bars to concrete in direct pullout conditions and found an average 75% bond strength for GFRP bars. Sharaky et al. [28] conducted an experimental study on critical factors affecting the bond strength of near-surface mounted GFRP and CFRP bars. El-Nemr et al. [14] investigated the bond-dependent coefficient, k_b , of GFRP and CFRP bars in normal- and high-strength concrete.

In the present study, a hybrid reinforcement bar of a steel core wrapped with a BFRP cover is proposed. Such reinforcement design overcomes the shortages of BFRP by combining the advantages of steel's high elastic modulus with BFRP's high tensile strength. Meanwhile, the proposed BFRP wrapping system may have the potential to solve or mitigate the problem of corrosion by wrapping the steel core with the non-corrosive BFRP cover. In addition, the overall cost can be reduced significantly by using a large amount of steel to replace expensive BFRP materials as the price of BFRP bars is much higher than that of steel rebar (approximate 1 USD and 0.5 USD per meter, respectively, for the same cross-sectional diameter of 10 mm).

A total of 11 specimens were prepared in this study, including 4 BFRP-steel hybrid ones and 7 GFRP bars. The tensile strength and

the elastic modulus of the proposed BFRP-steel reinforcement were experimentally investigated. As GFRP is the most commonly used FRP reinforcements, the test results are compared with steel and GFRP bars. In the meantime, the pull-out behaviours of the hybrid BFRP-steel bars reinforcement were experimentally investigated. Twelve specimens including 2 types of FRP materials (e.g., GFRP and BFRP) and 4 types of surface treatments (e.g., helically wrapped, sand coated, spirally wounded, and both spirally wounded & sand coated) were tested.

2. Experimental program

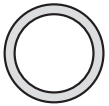
2.1. Specimen preparation of the tensile test

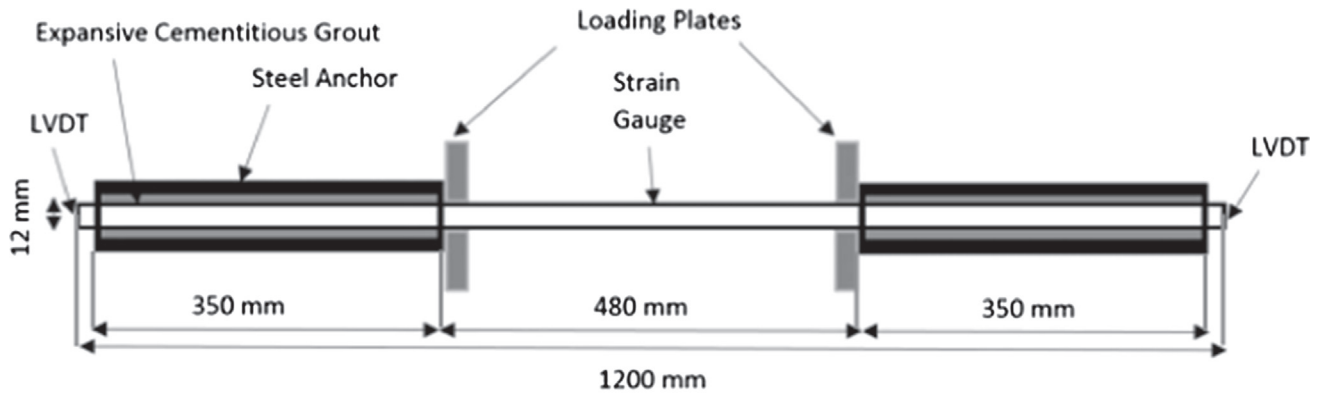
The tensile test and specimen preparation were conducted according to [4]. Seven GFRP specimens and four BFRP-steel hybrid bars were involved in the tensile test. Table 1 shows the bar designs of both types of bar specimens. The BFRP-steel hybrid bars were fabricated by a commercial manufacturer. The fabrication process includes pultrusion, braiding and surface coating. Both pultrusion and braiding processes of basalt fibres are applied to the steel core simultaneously, and epoxy is used as outside coating to solidify the composite bars and finalise the fabrication. Surface treatments are applied to improve the bond strengths of the bars and to ensure that pure tensile failure can be achieved.

Fig. 1 depicts an illustrative drawing of a detailed specimen design. As shown in Fig. 1, the total lengths of the GFRP and BFRP-steel bars were 1200 mm and 1240 mm, respectively. According to ASTM D7205 [4], the gauge length is required to be 40 times the nominal diameter of the bar. Therefore, the gauge lengths of the GFRP and BFRP-steel bars were determined to be 480 mm and 640 mm, respectively. Meanwhile, steel anchors were placed on both side of the specimens. According to ASTM D7205, the steel anchor length is required to be greater than 254 mm. Thus, the steel anchor length of GFRP bars was determined to be 350 mm, and the steel anchor length of BFRP-steel bars was 300 mm.

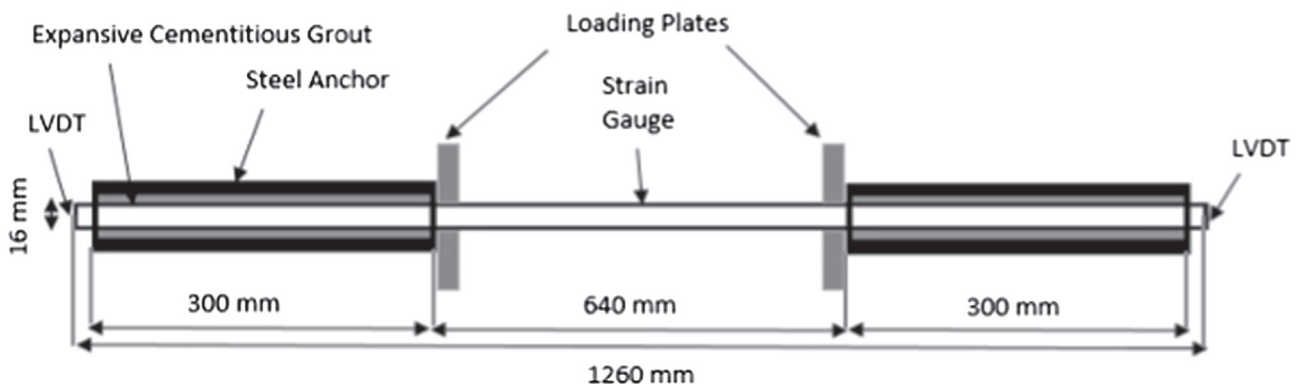
The steel anchors had an outer diameter of 42.4 mm with a wall thickness of 4 mm. The gap between the steel anchor and the specimen bar was filled by an expansive cementitious grout (Lanko 701 DuragROUT). The dry expansive cementitious grout had compressive strengths of approximately 67 MPa after 28 days curing. A concrete vibrator was used to reduce excess voids and to ensure the even distribution of the grout throughout the steel anchor. As shown in Fig. 2(a), wooden stoppers were placed at both ends of the steel anchors to ensure that the specimen bars were fixed at

Table 1
Specimen design for tensile test.

Bar Type	GFRP	BFRP-steel Hybrid
Diameter (mm)	12	16
Steel-FRP Cross-Sectional Area Percentages (%)	100% GFRP	56% Steel–44% Basalt
Cross-Sectional Diagram		
Cross-Sectional Image		



(a) Design of GFRP bars



(b) Design of BFRP-steel bars

Fig. 1. Specimen layout for tensile test.

(a) Wooden stoppers



(b) All prepared specimens

Fig. 2. Wooden stopper placement and all prepared specimens for tensile test.

the centre. The specimens were cured in a curing room for 28 days. Fig. 2(b) shows all the prepared specimens for tensile test.

2.2. Specimen preparation of pullout test

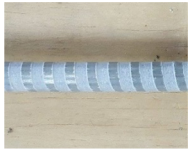
The bar specimens for the pullout experiment were both prepared and tested according to ACI Committee 440.3R-12 [1]. As shown in Table 2, there were a total of four groups of specimens, including helically wrapped GFRP bars, uniform sand-coated GFRP bars, spirally wounded BFRP-steel hybrid bars, and both spirally

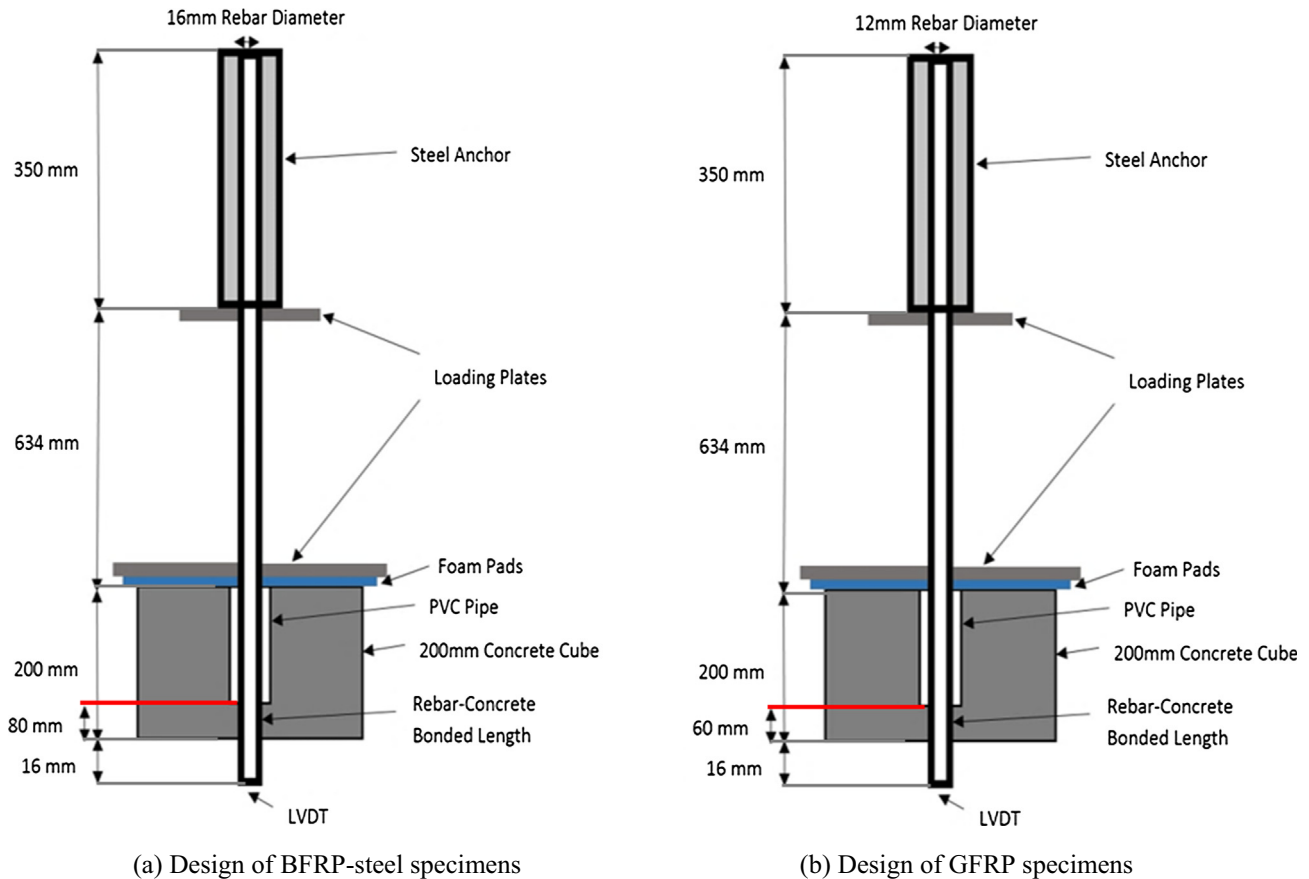
wounded & sand-coated BFRP-steel hybrid bars. Each group had three bar specimens with a total length of 1260 mm for GFRP bars and 1280 mm for BFRP-steel bars.

Fig. 3 shows illustrative diagrams of specimen designs for pullout test. A 350 mm steel anchor was placed at one end of the specimen with the same design philosophy as that used for tensile test. According to ACI 440.3R-12 [1], the length of the bar bonded to the actual concrete should be only five times the diameter, and thus, the rest of the bar embedded into concrete was bonded by a PVC pipe as shown in Fig. 3.

Table 2

Bar specimens with different surface treatments.

Bar Type	GFRP (helically wrapped)	GFRP (uniform sand coated)	BFRP-steel (spirally wounded)	BFRP-steel (spirally wounded/sand coated)
Diameter (mm)	12	12	16	16
Images of bar surface treatment				

**Fig. 3.** Specimen design for pullout test.

Twelve 200 mm cubes of timber moulds with a thickness of 16 mm were prepared to cast a concrete base. There were holes drilled at the centre of the bottom of the moulds with the same size of the bar's diameter. The holes were made in accordance with ACI 440.3R [1] to ensure that the bar end reached slightly outside the concrete block for easier laser LVDT placement. Extra 200 mm slabs with a centred hole that had the same size as the bar diameter were made to ensure that the bars were centred after the concrete was poured in.

Table 3 lists the concrete mixture of the base block. Twelve 100 × 200 mm cylinder concrete samples were cast to test the compressive strength of the concrete blocks. According to AS

1379–2007 [31], three cylindrical samples were tested every 7 days of a 28-day period. An average compressive strength of 35 MPa was observed from the samples cured for 28 days. Fig. 4 shows the final specimens prepared for the pullout tests.

2.3. Instrumentation and specimen testing

All the specimens were tested using an experimental apparatus in the concrete laboratory of the University of Western Australia. A Baldwin compression machine with a load capacity of 6000 kN and a maximum displacement of 500 mm was used for both tensile and pullout tests. The displacement of the hydraulic loading plate,

Table 3

Concrete mixture of the base block.

Component	Water	Cement	10 mm aggregates	7 mm aggregates	Minus 4 aggregates	Sand	f'_c MPa	Wet density
kg/m ³	205	426	444	306	130	843	35	2351



Fig. 4. Specimens prepared for pullout test.

strain gauge readings, and time were all recorded digitally. Important features, including the ultimate failure mechanisms of specimens, were photographed and recorded using a camera.

Fig. 5 indicates both tensile and pullout specimens set up in the Baldwin universal testing machine. Two of 200 mm steel loading plates with a thickness of 20 mm were placed between the loading and the stationary heads to transfer loads to the specimens. For the tensile tests, the loading rate for both GFRP and BFRP-steel bars was determined to be at 5 mm/min to satisfy the requirement of CSA standards [10]. As for the pullout tests, the loading rate was determined to be 1.3 mm/min based on ACI 440.3R [1].

For tensile tests, LVDTs were placed on both sides of the steel anchors to measure the slippage between the cementitious grout and the bar specimens under tensile loading, while the LVDT was placed on the free end of the bar specimens for the pullout test to test the slip of the bar displacement through the concrete block. The LVDT was also used to determine the slippage between the steel core and the BFRP layer.

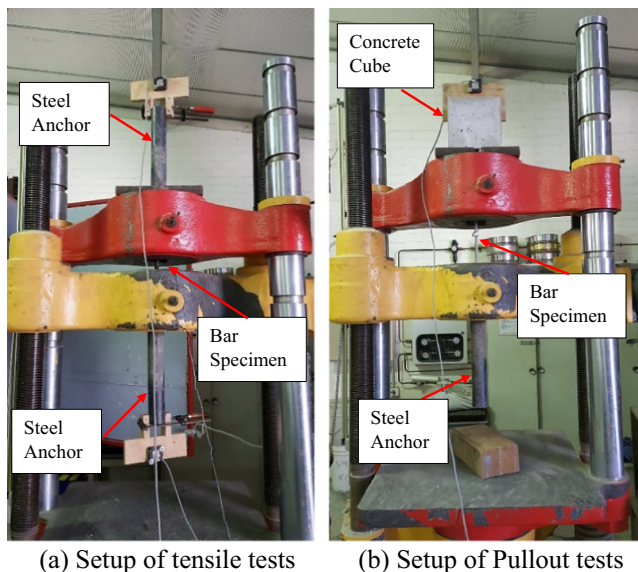


Fig. 5. Setup of both tests using a Baldwin machine.

3. Results and discussion

3.1. Calculations

The tensile strength, elastic modulus, and pullout strength were calculated based on the recorded data and the parameters of the specimens using the following equations. The tensile strength of each rebar was calculated using Eq. (1) according to CSA standards [10]:

$$f_u = \frac{P_{Max}}{A_{Rebar}} \quad (1)$$

where f_u is the tensile strength [Pa], P_{Max} is the maximum tensile load [N], and A_{Rebar} is the cross-sectional area of the bars [m²].

According to CSA standards [10], the elastic modulus was calculated using Eq. (2) below:

$$E_{Hybrid} = \frac{(P_1 - P_2)}{(\epsilon_1 - \epsilon_2)A_{Rebar}} \quad (2)$$

where E_{Hybrid} is the elastic modulus of the rebar [Pa]; P_1 & P_2 are the applied load at 50% and 25% of the ultimate load, respectively [N]; ϵ_1 & ϵ_2 are the strains at 50% and 25% of the ultimate load respectively; and A_{Rebar} is the cross-sectional area of the specimen [m²].

The pullout strength can be calculated using Eq. (3) in accordance with ACI 440.3R [1]:

$$\tau = \frac{F}{C_b * l} \quad (3)$$

where τ is the average bond stress [Pa], F is the tensile load [N], C_b is the equivalent circumference of the FRP bar [m], and l is the bonded length [m].

3.2. Results and discussion of tensile tests

Table 4 lists the average results of the tensile strength and the elastic modulus obtained from the tensile tests. Six GFRP and four BFRP-steel bars tested and recorded while one extra GFRP bar were used to investigate whether the expansive cementitious grout was strong enough to undergo the tensile tests. The tensile capacity of a normal steel rebar was also listed, and the improvement percentages were calculated and recorded in Table 4.

Table 4

Test result of tensile strength and Elastic Modulus.

	Steel	BFRP-steel	Improvement (%)	GFRP	BFRP-steel	Improvement (%)
Tensile Strength (MPa)	335	492.8	47	815.23	492.8	−40
Elastic Modulus (GPa)	200	129.17	−35	48.07	129.17	169



(a) GFRP bars



(b) BFRP-steel bars

Fig. 6. Failure of tensile tests.

(a) Debonding between sand coating and GFRP



(b) Debonding between steel bar and BFRP wrapping

Fig. 7. Debonding failures of specimens.

As shown in Table 4, both types of reinforcements improve the tensile strength of only the steel-reinforced concrete. The BFRP-steel bar reinforcement increased the tensile capacity by 47%, while the GFRP bar enhanced the tensile strength by around 2.5 times. However, the elastic modulus of the GFRP reinforcement only reached 24% of the steel rebar, while the proposed hybrid reinforcement could reach 65%, which means that the BFRP-steel

bars improved the elastic modulus by about 169% compared to the GFRP bars. Therefore, it can be concluded that the proposed BFRP steel reinforcements achieved a balance between tensile strength and elastic modulus, and it is able to enhance the tensile strength from steel reinforcement and improve the elastic modulus from GFRP reinforcement.

All failures of the specimens under tensile tests are shown in Fig. 6. For GFRP bars, sudden brittle failures with abruptly ruptured

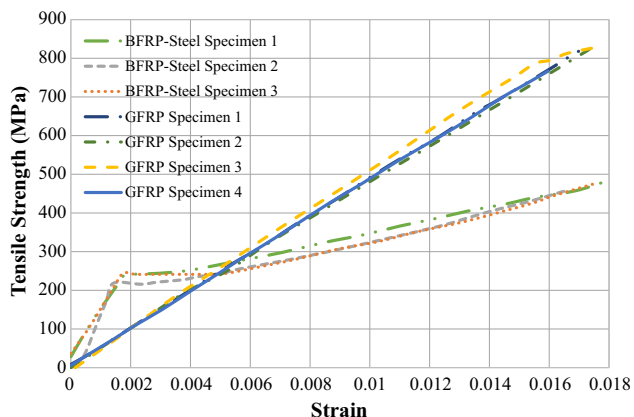
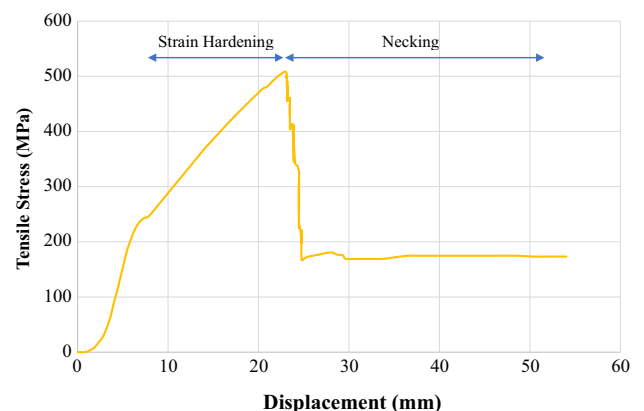
**Fig. 8.** Tensile strength versus strain curves for both the BFRP-steel and GFRP specimens.**Fig. 9.** Tensile stress vs displacement.

Table 5
Results of the pullout strength.

Pullout Strength (MPa)	BFRP-steel (Spiral Wounded)	BFRP-steel (Spiral Wounded/Sand Coated)	GFRP (Sand Coated)	GFRP (Helical Wrapped)
Sample 1	12.22	14.96	22.99 (excluded)	18.68 (excluded)
Sample 2	8.87 (excluded)	16.32	14.57	13.49
Sample 3	11.04	15.42	16.4	11.7
Average	11.63	15.57	15.49	12.59
CoV (%)	5.07	3.63	5.91	7.11

fibres were observed during the tests and confirmed by the stress-strain curves from Fig. 8, while more gradual failure modes occurred at BFRP-steel bars. As can be seen in Fig. 6, most of the specimens failed by fibre rupturing as the tensile loading exceeded the tensile strength of the tested bars. However, several failure cases caused by debonding were observed for both the GFRP and the BFRP-steel specimens. Fig. 7 indicates two example photos of debonding failures for the GFRP and the BFRP-steel specimens. For such kind of failure, the results were excluded from the calculation of average tensile strength as those values could not truly represent the maximum tensile strengths of the specimen bars.

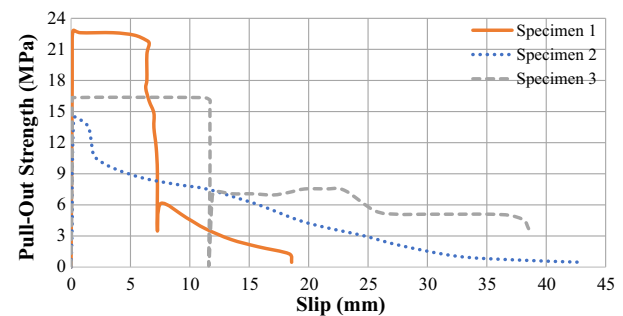
Fig. 8 indicates the stress-strain curves of all specimens excluding the ones failed by debonding behaviours. A linear stress-strain curve was observed from the GFRP specimens. The stress-strain curves of the hybrid BFRP-steel-reinforced concrete specimen showed a bilinear relationship, which was caused by the hybridization of two materials with a different elastic modulus and tensile strength. The steel dominated at a low tensile strength and yield at approximately 235 MPa at a strain of 0.002. Then the BFRP outer layer started to take all the loads after the steel failed and increased the tensile capacity to about 400 MPa.

Fig. 9 shows the tensile stress against displacement diagram of one typical BFRP-steel bar specimen. The strain hardening stage of steel is strengthened by a linear elastic behaviour due to the outside layer of basalt fibres. After the basalt reaches maximum tensile capacity and ruptures, the steel begins the necking process. However, with all the basalt-steel hybrid specimens, the fracturing process was not visualised. This phenomenon could be explained by the weak adhesive bond strength of the steel core and basalt layer in comparison to cementitious grout and basalt layer, which results in de-bonding, and thereby causes the steel core to slip through the basalt fibre. After 35 mm of machine load displacement, slippage of the steel core occurred as the LVDT laser on the steel core increased. These results can be beneficial as the slippage of the steel and basalt still provide continuous strength for the rebar.

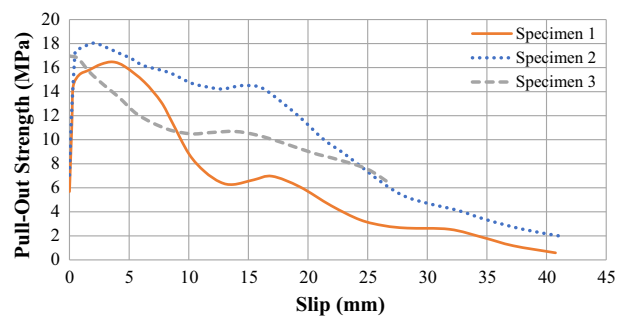
3.3. Results and discussion of pullout tests

Twelve specimens were tested to investigate the pullout behaviour of both the BFRP-steel and the GFRP bar reinforcements with different surface treatments. Table 5 indicates both individual and average pullout strengths and a coefficient of variation (CoV) of each group. When calculating the average strengths and CoVs, three results with significant deviations were excluded from calculations. The coefficient of variation (CoV) was calculated as standard deviation divided by the average value for each group. The test results were then assumed to be consistent as the CoV values of all cases were relatively small, less than 10%.

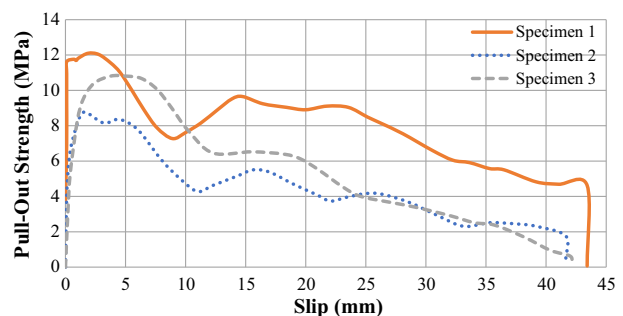
In Table 5, both sand-coated BFRP-steel and GFRP bars show the highest pullout strength (15.57 and 15.49 MPa, respectively), followed by the helically wrapped GFRP and the spiral wounded BFRP-steel bars (12.59 MPa and 11.63 MPa, respectively). The higher pull-out strengths of sand-coated bars is observed because the sand coating increases the surface roughness, which results



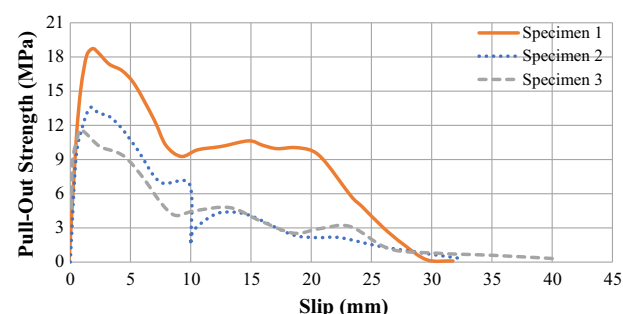
(a) Sand-coated GFRP specimens



(b) Spirally wounded/sand-coated BFRP-steel specimens



(c) Spirally wounded BFRP-steel specimens



(d) Helically wrapped GFRP specimens

Fig. 10. Pullout strength versus slip graphs for all specimens from four groups.

in a greater chemical bonding between the rod and the concrete. Consequently, an increased frictional resistance improved the pull-out strength.

Fig. 10 shows the detailed pullout strength versus the slip curve of each specimen. As can be seen from Fig. 10, all curves demon-

strate a similar stick-slip pattern, which is that, after reaching the maximum capacity, the pullout strength decreases suddenly at a fluctuating but continual rate until the bars are completely pulled out. Such stick-slip behaviour is caused by the surface roughness between the sliding bar and the concrete, and the



Fig. 11. Concrete blocks after pull-out test.



(a) Spirally wounded and sand-coated BFRP-steel bars



(b) Spirally wounded BFRP-steel bars



(c) Helically wrapped GFRP bars



(d) Sand-coated GFRP bars

Fig. 12. Demonstration of the bars after failure.

fluctuation of pullout strength indicates the change of surface characteristics and frictional conditions [30].

From Fig. 10(a) and (b), it can be seen that both sand-coated groups of bars had the highest bonding strength. However, different stress-slip behaviours can be observed.

Brittle failures can be found in the sand-coated GFRP group. The sand-coated GFRP bars showed good behaviour before reaching the peak bond stress. As can be seen in Fig. 10(a), significant vertical drops of pullout strength are seen for specimens 1 and 3. For specimen 2, a small near-vertical reduction can also be observed, though the drop is not as apparent as the other two specimens probably due to the lower peak pullout strength. This occurs because a sudden detachment of the interface between sand grains and the bar due to the less frictional resistance between the sand and the bars than that between the sand and the concretes.

The sand-coated BFRP-steel bars had more consistent behaviours among specimens. As shown in Fig. 10(b), a more gradual reduction of pull-out strength occurred possibly because of the hybridization of two surface treatments. It was observed that the spiral wounds reduced the sudden detachment of the sand coating and enabled a more gradual loss of bonding strength. In addition, the ultimate pull-out strength of such hybrid surface treatment was also observed to be slightly higher than that of uniformly sand-coated GFRP bars.

Comparing the two other surface treatments without sand coating, the helically wrapped GFRP specimens achieved a better bonding strength than the spirally wounded BFRP-steel bars. The bearing resistance may probably be affected by the factors of the shear strength of the bar-lug interface, lug dimensions, pitch of the lug, and number of lugs [30]. Therefore, the higher bond strength of the helically wrapped GFRP specimens could be caused by the larger size and larger number of lugs of the helically wrapping treatment than that of the spirally wounding treatment at the same bar length.

Figs. 11 and 12 shows the pull-out failure of the specimens. From Fig. 11, smooth surfaces are observed around the pull-out holes and no visual cracks can be found on the surfaces for all concrete cubes. Thus, it is assumed that no concrete failure occurred during the pull-out tests. Meanwhile, Fig. 12 shows that there is also no bar rupture happened for all bar specimens. Therefore, it is assumed that all bars should fail in pull-out failure mode only. From Fig. 12(a), the spiral indentations created discontinuities on the uniform sand-coated bar surface and prevented massive detachment of the sand coating, while massive detachment of the interface between sand coating and the GFRP bars can be observed from Fig. 12(d). Thus, the hybrid surface treatment is believed to provide a significant level of pull-out strength as well as a more ductile failure mode. From Fig. 12 (b) and (c), the larger gap size between lugs can be observed for helically wrapped GFRP bars, which may increase the surface roughness by allowing more concrete to be filled into the gaps, and consequently, improve the bonding behaviour.

4. Conclusion

The continual maintenance and inspection of steel-reinforced concrete can be costly, and a considerable amount of technical expense for repair may also be required. Therefore, this paper proposes a BFRP-steel hybridized bar reinforcement to provide a practical and feasible method for concrete in corrosive environments. Such hybridization protects the steel core with an anticorrosive basalt coating, and the low cost of steel improves the feasibility of using fibre reinforcement as the total material cost can be significantly reduced.

The proposed BFRP-steel bar can provide desirable mechanical properties of both tensile strength and elastic modulus from the

combination of steel and BFRP. The results indicate a 47% increase of tensile strength over the steel rebar as well as a 169% improvement of elastic modulus compared with GFRP bars, which proves that the proposed BFRP-steel bar enables a more balanced mechanical property which overcomes shortages of both steel's low tensile strength and basalt fibre's low elastic modulus.

Meanwhile, the bonding behaviour of the proposed bar reinforcement was also investigated based on four groups of surface treatments. Both sand-coated BFRP and GFRP specimen group showed the highest bonding strength. However, a brittle bonding failure was observed for sand-coated GFRP specimens because the sand coating detached abruptly from the bars. The spiral indentations on the sand-coated BFRP bar surface enabled a much more gradual failure. In addition, it can be observed from the other two groups that the larger gap size between lugs may induce a larger surface roughness and higher bonding strength.

Furthermore, although this study shows the potential application of using the proposed BFRP-steel bar for concrete reinforcement, more research should be conducted to investigate the long-term performance of the hybridized bars under different environmental conditions such as acidic and alkaline environments. In the meantime, further investigations of other mechanical properties, including transverse-shear strength, interlaminar shear strength, and flexural performance, can be valuable for future applications.

Conflict of interest

None.

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